MOHAMED R. SHOAIB

MACHINE LEARNING & DATA SCIENTIST ENGINEER

CONTACT DETAILS

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Nanyang Ave, Singapore

CAREER OBJECTIVES

Self-driven in work and a team player with good communication skills, seeking to join a good team. Interested in biomedical, data science fields and the Utilization of artificial intelligence techniques in health care applications and many other applications.

SKILLS

- Machine learning, Deep Learning, Transfer Learning, Object Detection, and Amazon Personalized Tools.
- Time Series and Forecasting.
- Computer Vision, Image Processing, NLP, Recommender System, and Reinforcement System.
- Embedded System Skills.
- Languages: C, Python, and MATLAB.
- Basics Problem Solving.
- Tensor Flow, Keras, Scikit-learn, Yolo, RNN, GAN, and Big data with Spark.
- Data visualization and Plotly and Dashboard.

EXPERIENCE

NTU (Nanyang Technological University, Singapore)

DOCTOR OF PHILOSOPHY at College of Computing and Data Science (CCDS) (FULL-TIME)

JAN 2023 - Present.

I am a full-time researcher working in machine learning, data science, and artificial intelligence, as well as working on applications like how to apply AI in biomedical, agricultural, and communication.

Shgardi Company

Machine Learning Engineer and Data Scientist Engineer.

FEB 2022 - Jan 2023.

I am working as a full-time machine learning engineer at Shgardi, which uses the Amazon personalized tool and time-series data for recommendation systems. In addition, working on fraud and fake user interactions.

EDUCATION

NTU (Nanyang Technological University, Singapore)

DOCTOR OF PHILOSOPHY at the College of Computing and Data Science (CCDS) at Nanyang Technological University (NTU), Singapore.

JAN 2023 - Present.

Information Technology Institute (ITI)

Diploma in Artificial Intelligence powered by EPITA School of Engineering and Computer Science.

APR 2021 - JAN 2022

Faculty of Engineering, Menoufia University.

Master's degree in Engineering Science.

Subject: Utilization of Artificial Intelligence Techniques in Healthcare Applications. (Pre-Master GPA: 3.46/4)

Oct 2019 - MARCH 2022

CERTIFICATIONS

Udacity Certified

AWS Machine Learning Foundations Nanodegree Program (2021)

Data Camp Certified

Data scientist, Data analyst, and Python programming Career Tracks. (2021)

IBM Certified

ML, Data Science, NLP, and AI courses. (2020)

Udemy Certified

ML, Data Science, NLP, and DL courses. (2019)

Machine Learning Specialization, Deep Learning Specialization

Master fundamental AI concepts and develop practical machine learning skills in the beginner-friendly, In addition, Master the fundamentals of deep learning and break into AI. Recently updated with cutting-edge techniques!

PROJECTS

Revolutionizing Global Food Security through AI Foundation Models

Oct 2023

- Objective: Addressing global food security challenges through data-driven solutions and integrated AI foundation models.
- Scope: Integrating AI models in various applications including crop type mapping, cropland mapping, field delineation, and crop yield prediction.
- Data Utilization: Leveraging multispectral imagery, meteorological data, soil properties, historical records, and high-resolution satellite imagery for comprehensive analysis.
- Advantages: Enhancing food security initiatives by providing accurate predictions, optimizing resource allocation, and facilitating informed decision-making.
- Transformational Impact: Serving as a powerful tool in overcoming limitations of current deep and machine learning methods, ultimately contributing to a sustainable and secure food future.

Unmasking Fraud in Food Delivery Business: Harnessing AWS' Random Cut Forest for Efficient Detection

Jul 2023

- Objective: Develop and evaluate a fraud detection model for identifying fraudulent cases and anomalous instances in a business system.
- Dilemma Resolution: Opted for a multivariate model (Random Cut Forest) over a univariate model (Isolation Forest) due to its ability to detect various types of anomalies and ensure system safety.
- **Model Effectiveness:** Successfully identified instances similar to known fraud cases and highlighted anomalies requiring further investigation, contributing to the development of a more robust supervised model.
- **Implications:** The findings from the investigations have the potential to enhance the system's accuracy in identifying future instances of fraud, thereby improving the overall security and trustworthiness of the food delivery business.

Utilization of Advanced Artificial Intelligence Tools for Critical Environmental Applications

May 2023

- ·Utilizing deep learning to overcome challenges in extracting useful information from plant images and satellite images.
- ·Combining data from various sources and sensor types to account for variations in lighting and sensor characteristics.
- Using image processing techniques and deep learning to develop smart agriculture applications.
- Using a spectral-spatial CNN to classify objects within the images .
- ·Evaluating and fine-tuning the models to improve performance.
- ·Deploying the trained models in a platform for satellite imagery analysis.
- ·Encryption and compression techniques will be used for more secure smart agriculture system.

WEB API for Time Series Forecasting

February 2022

- Deep learning models for Forecasting (LSTM).
- Data Preprocessing with Python.
- Web API for Production Implementation.
- Our goal is to predict the number of orders that can occur in a specific zone or location and time interval.

DL for Recommendation System

February 2022

- Amazon Personalize Tools.
- Deep learning models for Recommendation.
- Data Preprocessing with Python.
- Web API for Production Implementation.

DI for Brain Tumor Diagnosis

November 2021

- Deep learning models for classification.
- Different Image Processing Techniques: augmentation, Segmentation, and Super Resolution.
- Hardware Implementation with Web API.

Publications

Journals

1. Mohamed R. Shoaib et al, “Advanced Deep Learning Approaches for Fault Detection in Solar PV Systems: A Comparative Study of SPDA and AIFD-SolDL”, Accepted for publication in Applied Soft Computing on 11 Jul 2025.
2. Mohamed R. Shoaib et al, “Deep learning innovations in diagnosing diabetic retinopathy: The potential of transfer learning and the DiaCNN model”, published in Computers in Biology and Medicine on 11- Dec 2023. DOI: https://doi.org/10.1016/j.compbiomed.2023.107834.
3. Mohamed R. Shoaib et al, “Revolutionizing diabetic retinopathy diagnosis through advanced deep learning techniques: Harnessing the power of GAN model with transfer learning and the DiaGAN-CNN model”, Published in Biomedical Signal Processing and Control, Volume 99, 2025, https://doi.org/10.1016/j.bspc.2024.106790.
4. Mohamed R. Shoaib et al, “Improving brain tumor classification: An approach integrating pre-trained CNN models and machine learning algorithms”, Published in Heliyon.
5. Mohamed R. Shoaib et al. " Automated diabetic retinopathy classification using vision transformers on optical confocal microscopy images”, Accepted for publication in Applied Optics.
6. Mohamed R. Shoaib et al. "Efficient Framework for Brain Tumor Detection Using Different Deep Learning Techniques.", published in Frontiers in Public Health in 2022. DOI: 10.3389/fpubh.2022.959667. p.3374.
7. Mohamed R. Shoaib et al, "Simultaneous Super-Resolution and Classification of Lung Disease Scans." Diagnostics 2023, 13, 1319. https://doi.org/10.3390/diagnostics13071319.
8. Mohamed R. Shoaib et al, “Automatic modulation classification with 2D transforms and convolutional neural network”, published in Transactions on Emerging Telecommunications Technologies on 22 August 2022. DOI: https://doi.org/10.1002/ett.4623.
9. Shoaib, M.R. et al. Automated diagnosis of EEG abnormalities with different classification techniques. Med Biol Eng Comput (2023). https://doi.org/10.1007/s11517-023-02843-w.
10. Mohamed R. Shoaib et al, “Deep Convolutional Neural Networks for COVID-19 Automatic Diagnosis”, published in Microscopy Research and Technique on 31- Jan 2021. Nov 84, No. 11 (2021) pp.2504-2516. DOI: https://doi.org/10.1002/jemt.23713.
11. Mohamed R. Shoaib et al, “Efficient Deep Learning Models for Brain Tumor Detection with Segmentation and Data Augmentation Techniques.”, published in Concurrency and Computation: Practice and Experience on 08-Jan-2022. DOI: https://doi.org/10.1002/cpe.7031.
12. Mohamed R. Shoaib et al, “Hybrid Classification Structures for Automatic COVID-19 Detection”, published in Ambient Intelligence and Humanized Computing in 21- Dec 2021 pp, 1-16. DOI: 10.1007/s12652-021-03686-9.

Conferences:

1. Mohamed R. Shoaib et al, “AIFM6G: Exploring the Intersection of Artificial Intelligence Foundation Models and 6G Wireless Communication Networks”. Published in The IEEE 99th VTC2024 Singapore Conference.
2. Mohamed R. Shoaib et al, “Smart Dry Bean Classification: Unleashing AI-Powered Image Analysis for Superior Precision”. Published in The IEEE 101st VTC2024 Singapore Conference.
3. Mohamed R. Shoaib et al, “Advanced Anomaly Detection in PV Solar Cells: Leveraging Vision Transformers (ViT) and Machine Learning for Enhanced Fault Diagnosis”. Accepted for publication in The IEEE 102th VTC2024 Singapore Conference.
4. Mohamed R. Shoaib et al, “HE-GCN: A Homomorphic Encryption and Optics-Driven Graph Convolutional Network for Privacy-Preserving Fraud Detection”. Accepted for publication in The IEEE 102th VTC2024 Singapore Conference.
5. Mohamed R. Shoaiba, et al, P-YOLOv8: Efficient and Accurate Real-Time Detection of Distracted Driving. Accepted for publication in 28th Annual IEEE High Performance Extreme Computing Virtual Conference, Sep-2024.
6. Mohamed R. Shoaib et al, "Unmasking Fraud in Food Delivery Business: Harnessing Amazon Web Services’ Random Cut Forest for Efficient Detection," 2023 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME), Tenerife, Canary Islands, Spain, 2023, pp. 1-10, doi: 10.1109/ICECCME57830.2023.10252966.
7. Mohamed R. Shoaib et al, "Deepfakes, Misinformation, and Disinformation in the Era of Frontier AI, Generative AI, and Large AI Models," 2023 International Conference on Computer and Applications (ICCA), Cairo, Egypt, 2023, pp. 1-7, doi: 10.1109/ICCA59364.2023.10401723.
8. Mohamed R. Shoaib et al, "Detecting Fraud in Food Delivery: Leveraging Amazon Web Services' Random Cut Forest and XGBoost Models for Enhanced Security," 2024 International Conference on Computer and Applications (ICCA), Cairo, Egypt, 2024, pp. 1-6, doi: 10.1109/ICCA62237.2024.10927878.
9. Mohamed R. Shoaib et al, "Comprehensive Evaluation of Vision Transformer in MRI-based Brain Tumor Detection," 2024 International Conference on Computer and Applications (ICCA), Cairo, Egypt, 2024, pp. 1-6, doi: 10.1109/ICCA62237.2024.10927847.
10. Mohamed R. Shoaib et al, "A survey on the applications of frontier ai, foundation models, and large language models to intelligent transportation systems," 2023 International Conference on Computer and Applications (ICCA), Cairo, Egypt, 2023, pp. 1-6, doi: 10.1109/ICCA59364.2023.10401518.

CONFERENCES

1. Participated in the IEEE Vehicular Technology Conference (VTC) 3 times with 5 papers accepted for publication in VTC 99TH, 101st, and 102nd.
2. THE INTERNATIONAL CONFERENCE ON COMPUTER AND APPLICATIONS IEEE/ICCA’2023, November 28th – 30th 2023 - BUE - CAIRO – EGYPT Presented paper “Deepfakes, Misinformation, and Disinformation in the Era of Generative AI and Large AI Models”.
3. THE INTERNATIONAL CONFERENCE ON COMPUTER AND APPLICATIONS IEEE/ICCA’2023, November 28th – 30th 2023 - BUE - CAIRO – EGYPT Presented paper “A Survey on the Applications of Frontier AI, Foundation Models, and Large Language Models to Intelligent Transportation Systems”.
4. The 3rd International Conference on Electrical, Computer, Communications, and Mechatronics Engineering (ICECCME) IEEE Conference record number: 57830. Participate in this conference by submitting a paper and delivering a presentation about this paper in July 2023.
5. ICaTAS 2021: The 6th International Conference on Advanced Technology and Applied Sciences. Journal of Physics: Conference Series, Volume 2128, 12-14 October 2021, Cairo, Egypt. Participate in this conference by submitting a paper and delivering a presentation about this paper
6. The Port63 Challenge 2023 - Ideas Unlimited Finalist Team **(Team Golden Algorithm (Web 3 Track))**. Issued by NTU Innovation and Entrepreneurship · Sep 2023.